Nonlinear Blaschke Circle Dynamics: Complete Periodic Census and Stability

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Abstract

For the real family $B_a(z) = z(z-a)/(1-az)$, $0 \le a < 1$, we give a complete circle periodic-point census and its nonconstant stability multipliers. An explicit increasing lift proves that every positive iterate has exactly $2^n - 1$ fixed points, while off-circle contraction isolates the two attracting channels.

Keywords: Blaschke products; periodic orbits; stability multipliers; transfer operators; Fredholm determinants; singular limits.

中文摘要

本文研究实参数有限布拉施克圆周映射，证明所有迭代的不动点数量及本原 周期数量，并记录随轨道变化的稳定性乘子。严格递增的提升给出完整性， 圆内与圆外的收缩则隔离两个吸引通道。
关键词：布拉施克乘积；周期轨道；稳定性乘子；转移算子； 弗雷德霍姆行列式；奇异极限。

1 Object, provenance, and an all-period orbit theorem

The time variable is integer iteration of $\tau_a = B_a|_{\mathbb{T}}$, where $\mathbb{T} = \{z : |z| = 1\}$ and $0 \le a < 1$. All complex phases below retain their sign. The explicit transfer spectra of analytic circle maps are prior work of Slipantschuk, Bandtlow and Just [1], and the finite-Blaschke spectral theorem is due to Bandtlow, Just and Slipantschuk [2]. We give a direct real-family proof and a quantitative finite-section certificate. The nonlinear multiplier atlas differs from the workspace's linear expanding-circle and quadratic inverse-branch systems. This owner distinction is not a claim of literature priority.

Choose the lift $\psi_a(0) = 0$ of $B_a(e^{i\theta}) = e^{i\psi_a(\theta)}$. Logarithmic differentiation gives

$$\psi_a'(\theta) = 1 + \frac{1-a^2}{1-2a\cos\theta + a^2}, \qquad \frac{2}{1+a} \le \psi_a'(\theta) \le \frac{2}{1-a}. \tag{1}$$

The extrema occur at $\theta = \pi$ and 0. Thus every map in the chamber is expanding, orientation preserving and of degree two.

Theorem 1 (Complete orbit census)**.** *For every $n \ge 1$ the circle fixed count and primitive cycle count are*

$$F_n = 2^n - 1, \qquad P_n = \frac{1}{n}\sum_{d|n} \mu(n/d)(2^d - 1). \tag{2}$$

Every circle fixed point of every iterate is simple. For a primitive cycle γ of length p, its multiplier is $\Lambda_\gamma = \prod_{j=0}^{p-1} \psi_a'(\theta_j) > 1$; its sth repetition has length sp and multiplier Λ_γ^s.

Proof. The lift of the nth iterate satisfies $\psi_{a,n}(\theta + 2\pi) = \psi_{a,n}(\theta) + 2\pi 2^n$. The function $\psi_{a,n}(\theta) - \theta$ has positive derivative by (1) and increases by exactly $2\pi(2^n - 1)$ on a half-open fundamental interval. It meets that many multiples of 2π, each exactly once. Nonzero derivative of the fixed-point equation proves simplicity. The identity $F_n = \sum_{d|n} dP_d$ gives Moebius inversion; the chain rule proves the repetition law. □

Complex conjugation commutes with B_a. It pairs or fixes circle cycles, preserving their multipliers; it does not reverse time. Since the map is two-to-one, there is no globally defined inverse dynamics on the circle. The orientation is positive, and the real Perron weights have phase zero. Their multiplier dependence cannot be removed when constructing traces.

For $0 < s < 1$ the Blaschke factor satisfies

$$|B_a(z)| \le q_a(s) := s\frac{s+a}{1+as} < s \quad (|z| \le s). \tag{3}$$

The maximum modulus of $(z-a)/(1-az)$ on $|z| = s$ is $(s+a)/(1+as)$, as follows by differentiating its squared modulus with respect to $\cos\theta$. Therefore no nonzero interior point is periodic. The reflection identity $B_a(1/\bar{z}) = 1/\overline{B_a(z)}$ gives the same result outside the circle. The only off-circle fixed points of an iterate are 0 and ∞, with local multipliers $(-a)^n$.

The unweighted Artin–Mazur zeta already follows from (2):

$$\zeta_{\mathrm{AM}}(u) = \exp\sum_{n\ge 1}\frac{F_n u^n}{n} = \frac{1-u}{1-2u} \quad (|u| < 1/2). \tag{4}$$

It is independent of a, whereas the stability weights are not.
Round-zero certificate: complete orbit census, stability, repetitions, orientation, and off-circle separation.

References

[1] J. Slipantschuk, O. F. Bandtlow and W. Just, *Analytic expanding circle maps with explicit spectra*, arXiv:1306.0445 (2013). `https://arxiv.org/abs/1306.0445`.

[2] O. F. Bandtlow, W. Just and J. Slipantschuk, *Spectral structure of transfer operators for expanding circle maps*, Ann. Inst. H. Poincare C Anal. Non Lineaire **34** (2017), 31–43, Theorem 5.4. `https://doi.org/10.1016/j.anihpc.2015.08.004`.

[3] B. Simon, *Trace Ideals and Their Applications*, second edition, Mathematical Surveys and Monographs **120**, American Mathematical Society (2005). `https://bookstore.ams.org/SURV/120`.